SECTION 1 Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. DURABLE: (A) barely perceivable (B) easily destroyed (C) impudent (D) conclusive (E) outlandish

2. SLUMP: (A) upsurge (B) unification (C) clear view (D) desolate area (E) unexpected arrival

3. DISPUTE: (A) alleviate (B) affirm (C) enervate (D) answer caustically (E) inquire curiously

4. PROFANE: (A) happy (B) guarded (C) future (D) new (E) holy

5. RETROGRADE: (A) insolvent (B) cautious (C) spinning (D) moving forward (E) hindering success

6. DEBILITATE: (A) abbreviate (B) isolate (C) invigorate (D) infuriate (E) decorate

7. BALM: (A) residue (B) irritant (C) tinge (D) proper dose (E) weak solution

8. DEPLORE: (A) celebrate (B) disclose (C) intervene (D) accumulate (E) implore

9. PROPENSITY: (A) punctuality (B) aversion (C) ability to guess (D) lack of courage (E) grace in movement

10. DAUNT: (A) execute thoroughly (B) express carefully (C) hearten (D) debate (E) plead

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Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate
(C) sophisticated. .amateur (D) risqué. .crude
(E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Because management ---- the fact that employees find it difficult to work alertly at repetitious tasks, it sponsors numerous projects to ---- enthusiasm for the job.

(A) recognizes. .generate (B) disproves. .create
(C) respects. .quench (D) controls. .regulate
(E) surmises. .suspend

12. Eratosthenes' maps were ---- by the ---- state of exploration in his time, but they fit the facts then known.

(A) enhanced. .traditional
(B) disorganized. .stagnant
(C) limited. .primitive
(D) refuted. .flourishing
(E) determined. .exaggerated

13. Never forgetting his education as a preacher, Burwell often interrupts the narrative with ---- commentary, eager that his readers ---- the moral.

(A) romantic. .ignore (B) prophetic. .discredit
(C) bombastic. .disregard (D) apathetic. .heed
(E) didactic. .grasp

14. The training center, clean and regimented, is ---- to those seeking the ---- once associated with boxing.

(A) convincing. .chaos
(B) disappointing. .seediness
(C) surprising. .austerity
(D) refreshing. .camaraderie
(E) inspiring. .ambition

15. Of ---- disposition, she spoke very little, even when in the company of her good friends.

(A) an amiable (B) a benign (C) a carping
(D) a taciturn (E) an inhumane

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. CLOTH : MILDEW :: (A) wood : timber
(B) insect : termite (C) flower : mushroom
(D) iron : rust (E) antidote : poison

17. CRITIC : JUDGMENT :: (A) satirist : ridicule
(B) doctor : rejection (C) writer : relaxation
(D) miser : generosity (E) exile : patriotism

18. ZOO : ANIMAL :: (A) safari : guide
(B) supermarket : shopper (C) herd : cattle
(D) river : stream (E) museum : painting

19. FATHOM : DEPTH :: (A) mile : hour
(B) volume : space (C) porthole : window
(D) knot : speed (E) port : starboard

20. OFFHAND : FORETHOUGHT ::
(A) furtive : secrecy (B) voluntary : coercion
(C) ingenious : talent (D) artistic : success
(E) fanciful : elaboration

21. OPAQUE : LIGHT ::
(A) dense : weight (B) buoyant : air
(C) hot : radiation (D) watertight : dampness
(E) brilliant : illumination

22. AMORPHOUS : SHAPE :: (A) obvious : evidence
(B) humble : belief (C) nondescript : classification
(D) momentary : fame (E) ambiguous : obscurity

23. SYBARITE : LUXURY ::
(A) hermit : seclusion
(B) student : school
(C) employee : office
(D) mathematician : complexity
(E) official : expectation

24. ANNEX : BUILDING :: (A) addendum : book
(B) scenario : play (C) program : computer
(D) landscaping : park (E) duplex : apartment

25. BOAST : SPEAK :: (A) shout : utter
(B) blare : hear (C) stumble : fall
(D) stare : see (E) swagger : walk

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

From the beginning, this trip to the high plateaus in Utah has had the feel of a last visit. We are getting beyond the age when we can unroll our sleeping bags under any pine or in any wash, and the gasoline situation throws the future of automobile touring into doubt. I would hate to have missed the extravagant personal liberty that wheels and cheap gasoline gave us, but I will not mourn its passing. It was part of our time of wastefulness and excess. Increasingly, we will have to earn our admission to this spectacular country. We will have to come by bus, as foreign tourists do, and at the end of the bus line use our legs. And if that reduces the number of people who benefit every year, the benefit will be qualitatively greater, for what most recommends the plateaus and their intervening deserts is not people, but space, emptiness, silence, awe.

I could make a suggestion to the road builders, too. The experience of driving the Aquarius Plateau on pavement is nothing like so satisfying as the old experience of driving it on rocky, rutted, chuckholed, ten-mile-an-hour dirt. The road will be a lesser thing when it is paved all the way, and so will the road over the Fish Lake Hightop, and the one over the Wasatch Plateau, and the steep road over the Tushar, the highest of the plateaus, which we will travel tomorrow. To substitute comfort and ease for real experience is too American a habit to last. It is when we feel the earth rough to all our length, as in Robert Frost's poem, that we know it as its creatures ought to know it.

26. According to the author, what will happen if fewer people visit the high country each year?

(A) The characteristic mood of the plateaus will be tragically altered.
(B) The doctrine of personal liberty will be seriously undermined.
(C) The pleasure of those who do go will be heightened.
(D) The people who visit the plateaus will have to spend more for the trip.
(E) The paving of the roads will be slowed down considerably.

27. The author most probably paraphrases part of a Robert Frost poem in order to

(A) lament past mistakes
(B) warn future generations
(C) reinforce his own sentiments
(D) show how poetry enhances civilization
(E) emphasize the complexity of the theme

28. It can be inferred from the passage that the author regards the paving of the plateau roads as

(A) a project that will never be completed
(B) a conscious attempt to destroy scenic beauty
(C) an illegal action
(D) an inexplicable decision
(E) an unfortunate change

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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Sergeant Blake was a man to whom memories were an encumbrance and anticipations a superfluity. That projection of consciousness into days gone by and to come, which makes the past a synonym for the pathetic and the future a word for circumspection, was foreign to Blake. With him the past was yesterday; the future, tomorrow.

Line
(5)

On this account he might, in certain lights, have been regarded as fortunate. For it may be argued with great plausibility that reminiscence is less an endowment than a disease, and that expectation in its only comfortable form—that of absolute faith—is practically an impossibility; whilst in the form of hope and the secondary compounds, patience, impatience, resolve, curiosity, it is a constant fluctuation between pleasure and pain.

(10)

(15)

Sergeant Blake, being entirely innocent of the practice of expectation, was never disappointed. To set against this negative gain there may have been some positive losses from a certain narrowing of the higher tastes and sensations which it entailed. But limitation of the capacity is never recognized as a loss by the loser therefrom: in this attribute, moral or aesthetic poverty contrasts plausibly with material poverty, since those who suffer do not mind it, whilst those who mind it soon cease to suffer. What Blake had never enjoyed, he did not miss.

(20)

(25)

29. The passage is primarily concerned with

(A) recounting Blake's past
(B) evaluating Blake's personality
(C) showing Blake's interaction with other characters
(D) pondering Blake's future behavior
(E) justifying the author's philosophy of life

30. The author's statement that Sergeant Blake could be "regarded as fortunate" (line 9) is

(A) a sincere expression of esteem for Blake's character
(B) an ironic recognition of the advantages of Blake's limitations
(C) an obviously false statement intended to amuse
(D) a contradiction of what the author later says about Blake
(E) an indication of the author's belief in the power of fate

31. The author states that thinking about the future is an activity that

(A) distinguishes humans from animals
(B) is restricted to those with absolute faith
(C) has no beneficial effects
(D) results in moral or aesthetic poverty
(E) subjects one to emotional turmoil

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Literary study in the United States has reflected the taste of the majority while leaving unturned the forgotten pages of our national literature. The intellectual flexibility inherent in a multicultural nation has been stifled in classrooms where emphasis on British-American literature has not reflected the cultural amalgam of our country.

The literary disenfranchisement of particular groups is evidenced by the lack of serious study given to oral literature. American oral literature is characterized by its stylistic freshness relative to more commonly studied forms and by the variety of its metaphors and choice of subject. The corn grinding song of the Navajo is typical. Of course, when languages have widely dissimilar structures, translations are rarely faithful. Yet the picture of a rainbow moving across the plains to the waiting corn and, by symbolic correlation, of humanity's utter dependence upon rain is vivid even in translation. Rain is a positive metaphor among the Navajo: a welcome guest, for example, is often compared to a thundershower. Some of the experiential complexities which seem inexpressible in English are less so in other American languages which are not limited by spatial and temporal priorities of syntax. Without using extensive description, the Navajo poet created an epiphany, intense and fleeting, much like the Japanese haiku.

Black slaves also had a moving oral tradition: consider the poetry of the spirituals and animal fables, the latter usually conveying a lesson in survival in a hostile environment. Furthermore, Yiddish stories once told in the ghettos of Europe provided inspiration for such contemporary American writers as Bellow and Malamud, yet these stories are largely ignored because they are untranslated.

The common argument that the nature of an original is obscured by translation has little validity. An original translation introduces a new work. If the translation is of high quality, it is relatively unimportant whether or not it is an exact duplicate. Moreover, translations provide the vitality of heterogeneous literary influences for aspiring writers. Schoolcraft's compendium of Chippewa poetry resolves the translator's dilemma by presenting two translations of each song. The first is a literal presentation, the second captures the "feel" of the original. This technique is not practical for longer works, however.

Traditional forms of verse—epic, pastoral, metrical romance—do not include the Chippewa medicine songs, war songs, and dream songs. This is not to argue that these categories should supplant traditional genres but that traditional literary modes do not exhaust human artistic potential.

32. Which of the following titles best describes the content of the passage?

(A) Who Is to Blame for the Intellectual Rigidity of American Classrooms?
(B) Wellsprings of Talent in Contemporary American Literature
(C) The Need to Reconsider Traditional Verse Forms
(D) A Plea for the Neglected Pages of Our Literature
(E) The Translator's Dilemma

33. According to the author, those who criticize the value of literature in translation do so on the grounds that

(A) oral literature cannot be subjected to literary analysis
(B) translation is not practical except in short poems and stories
(C) the feeling of the original is often distorted
(D) traditional works are more generally appealing
(E) English speakers will not appreciate works translated from languages of dissimilar structure

34. The author implies that the traditional study of verse forms should be

(A) temporarily relegated to a minor position in favor of fresher forms
(B) supplanted by vital new modes of native American poetry
(C) modernized for use in contemporary classrooms
(D) supplemented by other forms from American literature
(E) subjected to exhaustive categorization so that it will be more meaningful

35. The author cites Bellow and Malamud as examples of authors who

(A) were inspired by an oral tradition
(B) overcame the indifference of American audiences
(C) are a source of motivation to young writers
(D) are not widely translated
(E) convey lessons in survival in a hostile world



36. What the author says about Schoolcraft's technique in presenting Chippewa poetry in translation suggests which of the following situations?

(A) Depicting a scene by presenting a photograph, then an artist's rendering
(B) Writing a novel in which characters' desires are contrasted with actual events
(C) Presenting a concert in which a classical piece is played by a marching band
(D) Presenting a drama first as a stage play, then as a motion picture
(E) Illustrating a magazine article with caricatures accompanied by descriptive captions

---

In 1816, I was consulted by a young woman presenting general symptoms of disease of the heart. Owing to her stoutness, little information could be
Line gathered by application of the hand and percussion.
(5) The patient's age and sex did not permit me to resort to the kind of examination I have described earlier [that is, direct application of the ear to the chest]. I recalled a well-known acoustic phenomenon: namely, if you place your ear against one end of a
(10) wooden beam the scratch of a pin at the other extremity is most distinctly audible. It occurred to me that this physical property might serve a useful purpose in the case with which I was then dealing. Taking a sheaf of paper I rolled it into a very tight
(15) roll, one end of which I placed over the precordial region, whilst I put my ear to the other. I was both surprised and gratified at being able to hear the beating of the heart with much greater clearness and distinctness than I had ever done before by direct
(20) application of my ear.

I saw at once that this means might become a useful method for studying, not only the beating of the heart, but likewise all movements capable of producing sound in the thoracic cavity, and that
(25) consequently it might serve for the investigation of respiration, the voice, rales, and even possibly the movements of a liquid effused into the pleural cavity or pericardium.

37. It can be inferred from the passage that the patient's stoutness was a handicap to the author because the stoutness

(A) inhibited the production of sound when the chest was tapped in a prescribed manner
(B) prevented the use of pressure to find tumors and other growths in the chest
(C) caused wheezing and other chest noises that interfered with diagnosis
(D) precluded the use of more strenuous methods of diagnosis
(E) muffled the sounds of the chest so that they could not be detected by an ear placed on the chest

38. The use of the sheaf of paper as described in lines 14-16 had which of the following immediate effects?

(A) It intensified the author's concern for the patient.
(B) It allowed the diagnosis of irregular heartbeats.
(C) It opened the way for research into medical applications of wood and paper.
(D) It amplified the sounds the author wished to hear.
(E) It allowed the author to analyze the patient's respiration rate.

39. The author's tone is best described as

(A) ecstatic (B) compassionate (C) arrogant (D) apprehensive (E) objective

40. The main purpose of the passage is to

(A) characterize the inventiveness of nineteenth-century physicians
(B) describe how a medical instrument was first devised
(C) explain an unusual cure for a common ailment
(D) call for investigation of the heart and lungs
(E) reveal the difficulties of early medicine

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 3 Time—30 minutes 25 Questions

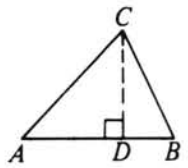
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $k = -1$, then $k + k^2 + k^3 + k^4 =$

   (A) -4 (B) -2 (C) 0 (D) 2 (E) 4

2. At 9:00 a.m. the temperature of a solution is -20°C. This temperature rises at a uniform rate and at noon the reading is 0°C. What will be the temperature of the solution at 3:00 p.m. on the same day, if it continues to rise at the same rate?

   (A) -20°C
   (B) -10°C
   (C) 0°C
   (D) 10°C
   (E) 20°C

3. If the sum of x and $x + 1$ is greater than 7, which of the following could be a value of x?

   (A) -4
   (B) 0
   (C) 1
   (D) 3
   (E) 4

Questions 4-5 refer to the following definition.

For all positive numbers p, where $p \neq 1$, define p^* by the equation $p^* = \dfrac{p + 2}{p - 1}$.

4. If $p = 4$, then $p^* =$

   (A) $\frac{6}{5}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) 2 (E) 3

5. If p is positive and $p \neq 1$, then $\frac{1}{2}\left(\frac{1}{p^*}\right)$ is equal to which of the following?

   (A) $\dfrac{p - 2}{p + 1}$

   (B) $\dfrac{p - 1}{2(p + 2)}$

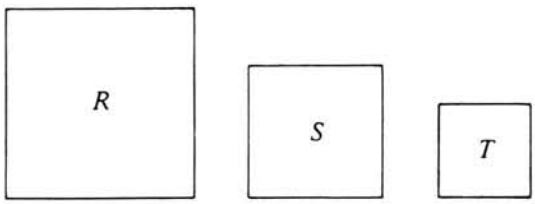
   (C) $\dfrac{2(p - 1)}{p + 2}$

   (D) $\dfrac{p + 2}{p - 1}$

   (E) $\dfrac{2(p + 2)}{p - 1}$

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6. Three squares R, S, and T are shown above. The area of R is twice the area of S, and the area of S is twice the area of T. If the area of S is 1, what is the sum of the areas of all three squares?

(A) $2\frac{1}{2}$

(B) $3\frac{1}{2}$

(C) 4

(D) $5\frac{1}{4}$

(E) 6

---

7. If $40{,}404 + x = 44{,}444$, then $40{,}404 - 10x =$

(A) −4.04
(B) 0
(C) 4
(D) 4.04
(E) 40.4

Questions 8-9 refer to the following definition.

For any positive integer k, let $\boxed{k}$ represent the greatest odd integer that divides k; for example $\boxed{36} = 9$.

8. $\boxed{30} =$

(A) 5 (B) 6 (C) 10 (D) 15 (E) 25

9. Which of the following is equal to $\boxed{2^9}$?

(A) 1 (B) 2 (C) 2^3 (D) 2^6 (E) 2^9

10. Several people are standing in a straight line. Starting at one end of the line, Bill is counted as the 5th person and, starting at the other end, he is counted as the 12th person. How many people are in the line?

(A) 15
(B) 16
(C) 17
(D) 18
(E) 19

11. If $\frac{x}{y}$ is a fraction greater than 1, which of the following must be less than 1 ?

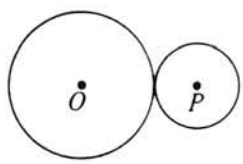
(A) $\frac{2y}{x}$

(B) $\frac{x}{2y}$

(C) $\sqrt{\frac{x}{y}}$

(D) $\frac{y}{x}$

(E) $\left(\frac{x}{y}\right)^2$



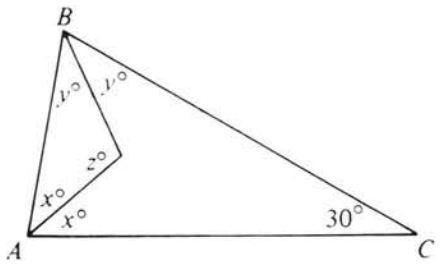
12. In the figure above, the circles with centers O and P have radii of 5 and 3, respectively, and have exactly one point in common. If a circle C (not shown) is drawn such that O is its center and P is on its circumference, what is the radius of circle C ?

(A) 8 (B) 10 (C) 11 (D) 16
(E) It cannot be determined from the information given.

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13. $\left[(2x^2y^3)^2\right]^3 =$

(A) $4x^4y^6$ (B) $12x^4y^6$ (C) $64x^4y^6$
(D) $64x^{12}y^{18}$ (E) $64x^{64}y^{216}$



14. In the figure above, if $\angle BAC = 80°$, then $z =$

(A) 100
(B) 105
(C) 110
(D) 120
(E) 125

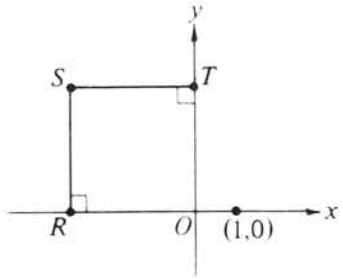
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15. If $\left(\frac{2}{10}\right)^3$ is equal to q thousandths, what is the value of q ?

(A) 5 (B) 6 (C) 8 (D) 23 (E) 32

16. If Town A is 300 kilometers due east of Town B and Town C is 400 kilometers due south of Town B, then the shortest distance, in kilometers, from Town A to Town C is

(A) 450
(B) 500
(C) 550
(D) 600
(E) 700



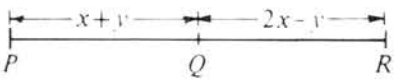
17. In the figure above, $RS = ST$ and the coordinates of S are $(k, 3)$. What is the value of k ?

(A) -3 (B) $-\sqrt{3}$ (C) 0 (D) $\sqrt{3}$ (E) 3

---

18. Car I averages 30 miles per gallon of gasoline while car II averages 25 miles per gallon. If gasoline costs \$2 per gallon, how much more would the gasoline cost for a 600-mile trip in car II than in car I ?

(A) \$16
(B) \$12
(C) \$10
(D) \$8
(E) \$6



19. Segment PR is divided into two segments with lengths as shown above. If Q is the midpoint of PR, which of the following statements must be true?

(A) $x = 0$ (B) $x = \frac{1}{2}y$ (C) $x = \frac{2}{3}y$
(D) $x = y$ (E) $x = 2y$

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20. What is the perimeter of a square that has the same area as a circle that has circumference 1 ?

(A) $4\sqrt{\pi}$

(B) $\frac{2}{\sqrt{\pi}}$

(C) $\frac{1}{2\sqrt{\pi}}$

(D) $\frac{1}{\pi}$

(E) 1

21. The average (arithmetic mean) of n numbers is 10 and the average of 5 of these numbers is 8. In terms of n, what is the average of the remaining numbers?

(A) $\frac{10n + 40}{5}$

(B) $\frac{10n - 40}{5}$

(C) $\frac{40 - 10n}{5}$

(D) $\frac{40 - 10n}{n - 5}$

(E) $\frac{10n - 40}{n - 5}$

22. A train started at station S and has been traveling r miles per hour for t hours. In terms of r and t, how many miles from station S had the train traveled $\frac{3t}{4}$ hours ago?

(A) $\frac{rt}{12}$

(B) $\frac{3rt}{4}$

(C) rt

(D) $\frac{4rt}{3}$

(E) $\frac{rt}{4}$

23. If $2x + y = 15$ and $x + 2y = -3$, then $2x + 2y =$

(A) 8
(B) 12
(C) 18
(D) 24
(E) 36

24. If r is a positive integer and the ratio $\frac{r}{s}$ is $\frac{1}{3}$, which of the following could be the value of $\frac{r^2}{s}$?

I. $\frac{1}{3}$

II. 1

III. $\frac{4}{3}$

(A) None
(B) I only
(C) I and II only
(D) I and III only
(E) I, II, and III

25. What is the number of different pairs of parallel edges on a cubical wooden block?

(A) 18 (B) 12 (C) 8 (D) 6 (E) 4

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SECTION 5 Time—30 minutes 35 Questions

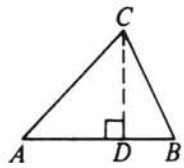
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The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $x = m - n$ and $y = n - m$, what is $x - y$ when $m = 6$ and $n = 6$?

(A) -12 (B) 0 (C) 6 (D) 12 (E) 24

Questions 2-3 refer to the following price list.

Number of Donuts	Total Price
1	\$0.40
Box of 6	\$1.89
Box of 12	\$3.59

2. Of the following, which is the closest approximation of the cost per donut when one purchases a box of 6?

(A) \$0.20 (B) \$0.30 (C) \$0.40
(D) \$0.50 (E) \$0.60

3. What would be the least amount of money needed to purchase exactly 21 donuts?

(A) \$5.88
(B) \$6.68
(C) \$7.19
(D) \$7.38
(E) \$8.40

4. If x and y are positive integers and $x + y = 12$, what is the greatest possible value of xy?

(A) 11 (B) 12 (C) 32 (D) 36 (E) 144

$$\begin{array}{r} R\ S \\ +\ S\ R \\ \hline 5\ 5 \end{array}$$

5. In the addition problem above, $0 < S < 5$ and $0 < R < 5$. How many different integer values of S are possible?

(A) One
(B) Two
(C) Three
(D) Four
(E) Five

6. $\sqrt{\frac{\sqrt{16}}{4}} =$

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$ (E) 4

7. If $x \neq 0$ and $3x^2 - 12x = 0$, what is the value of x?

(A) -9
(B) -4
(C) 3
(D) 4
(E) 9

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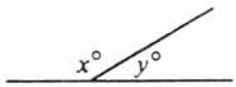
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	2^{16}	1^{32}
9.	$3.7 + 2.6 + 7$	$\frac{73.7 + 72.6}{10}$

A fair coin is tossed 2 times.

10.	The chances of getting 2 heads	The chances of getting no heads

$5t > 3s$

11.	t	s

In the last step of an arithmetic problem, John added 15 to the value he had obtained so far, when he should have subtracted 30. He made no other mistakes.

12.	The correct answer to the problem	John's answer to the problem
13.	4×5^{13}	5×4^{13}

$\frac{x}{y} = \frac{1}{6}$

14.	y	$6x$

	Column A	Column B
	x is an integer.	
15.	The greatest possible value of $4 - x^2$	5

x, y, and z are equal to 1, 2, and 3, but not necessarily in that order. $y > x$

16.	$y + z - x$	2

The number of employees in department J is 3 less than 5 times the number of employees in department P. There are n employees in department P.

17.	The number of employees in department J	$5n - 3$

$x^2 - 13x + 42 = 0$

18.	x	8

The list price of a watch is x dollars.

19.	The price of the watch at 25 percent off the list price	$0.75x$ dollars

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

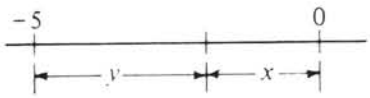
Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

	Column A	Column B
20.	$\dfrac{1}{\frac{10}{7} - 1}$	2

$$\frac{7}{8} < x < \frac{8}{7}$$

21.	x	$\frac{10}{9}$

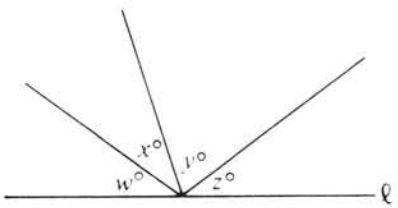


x and y are lengths of segments on the number line above.

22.	x	$5 - y$

Polygon X has equal sides and equal angles. The perimeter of polygon X is 10.

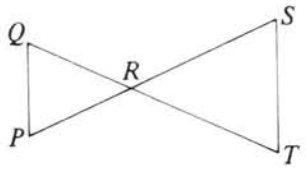
23.	The number of sides of polygon X	5



$w = x = z$ and $\frac{y}{z} = \frac{2}{1}$

24.	y	70

	Column A	Column B



Note: Figure not drawn to scale.

Line segments QT and PS intersect at point R.
$PQ \parallel ST$

25.	The length of QR	The length of PR

$$x > y > 0$$

26.	$x^2 - y^2$	$x^2 - y$

10 millimeters = 1 centimeter

27.	The area of a square that measures 5 centimeters on a side	10 times the area of a square that measures 5 millimeters on a side

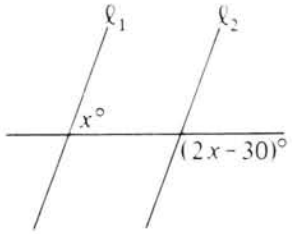
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5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. Which of the following equals the ratio of $2\frac{1}{2}$ days to 7 days?

(A) 2:35
(B) 2:9
(C) 2:7
(D) 5:14
(E) 5:9



29. In the figure above, if $\ell_1 \parallel \ell_2$, then $x =$

(A) 30 (B) 50 (C) 70 (D) 105 (E) 210

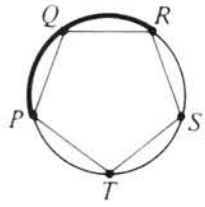
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30. If $(1{,}000+6)^2 = 10^6 + n$, then $n =$

(A) 36
(B) 6,036
(C) 12,006
(D) 12,036
(E) 60,036

31. If the sum of two angles of a parallelogram is 100°, what is the average (arithmetic mean) of the measures of the other two angles?

(A) 50°
(B) 90°
(C) 130°
(D) 180°
(E) 260°



32. In the figure above, inscribed polygon $PQRST$ is equilateral. If the diameter of the circle is 10, then the length of arc PQR is

(A) 10π (B) 4π (C) 2π (D) $\frac{4\pi}{5}$ (E) $\frac{2\pi}{5}$

---

33. According to the formula $F = \frac{9}{5}C + 32$, if the Fahrenheit (F) temperature increased 27°, by how many degrees would the Celsius (C) temperature be increased?

(A) 9° (B) 15° (C) 47°
(D) $48\frac{3}{5}°$ (E) 59°

34. A square and an equilateral triangle have perimeters S and T, respectively. If s and t are the respective lengths of a side of the square and a side of the triangle, then, in terms of their perimeters, $s - t =$

(A) $\frac{S-T}{7}$
(B) $\frac{4T-3S}{7}$
(C) $\frac{3S-4T}{7}$
(D) $\frac{4T-3S}{12}$
(E) $\frac{3S-4T}{12}$

35. If 3 percent of $(x+y)$ is 12 and x is a negative integer, what is the least possible value of y?

(A) 5 (B) 41 (C) 399 (D) 400 (E) 401

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 6 Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. DISMANTLE: (A) recall (B) approve (C) rebuild (D) embrace (E) dwell

2. HIDEOUS: (A) careful (B) economical (C) independent (D) delightful (E) assenting

3. COMPLEX: (A) elementary (B) distinct (C) precise (D) immature (E) reasonable

4. ABUSIVENESS: (A) retaliation (B) perception (C) estrangement (D) cordiality (E) exclusivity

5. INOPPORTUNE: (A) optional (B) stratified (C) methodical (D) insistent (E) convenient

6. EMBED: (A) cool down (B) keep pure (C) energize (D) exalt (E) extract

7. CORRUGATED: (A) narrow (B) heavy (C) smooth (D) firm (E) split

8. MORBIDITY: (A) fervor (B) meditation (C) indifference (D) fearlessness (E) wholesomeness

9. ILLUSTRIOUS: (A) abstract (B) idle (C) ill-tempered (D) infamous (E) persistent

10. VOLUMINOUS: (A) indistinct (B) slight (C) inaudible (D) courteous (E) misconceived

11. EQUANIMITY: (A) dedication (B) unselfishness (C) innocence (D) lack of composure (E) lack of success

12. INCONSEQUENTIAL: (A) transitional (B) natural (C) flawless (D) lively (E) pivotal

13. CONSONANCE: (A) contempt (B) discord (C) coarseness (D) extreme uncertainty (E) mindless repetition

14. VERBOSE: (A) energetic (B) relaxed (C) exotic (D) complimentary (E) terse

15. BURGEON: (A) vary (B) wither (C) move freely (D) leave alone (E) act cautiously

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. . respectable (B) extensive. . moderate (C) sophisticated. . amateur (D) risqué. . crude (E) perfect. . spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Although the shock of the accident had temporarily hindered his ability to think rationally, his instincts and sense of morality somehow seemed ----.

(A) intact (B) affected (C) unreachable (D) inconsistent (E) unresponsive

17. Metabolic by-products of oxygen are so ---- that breathing organisms have developed elaborate defense systems against them.

(A) insignificant (B) useful (C) toxic (D) unpredictable (E) necessary

18. Before the 1930's, the knowledge one could give children would last them their working lives, whereas today the technological revolution often renders familiar scientific developments ---- overnight.

(A) valid (B) expedient (C) obsolete (D) immortal (E) memorable

19. Ideally, every election ought to be ---- by a majority of ---- citizens, but it sometimes happens that the outcome is determined by those people who know least about politics.

(A) revoked. .esteemed (B) proposed. .prominent (C) regulated. .local (D) conducted. .ordinary (E) decided. .informed

20. She is one of the most ---- public figures in the country today, disarmingly candid and seemingly fearless of any consequences.

(A) prudent (B) agreeable (C) outspoken (D) indecisive (E) discerning

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6

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

During the 1840's, some doubts were cast on the applications of Newton's theory of gravitation to problems of planetary motion when it was found that the planet Uranus was not following the orbit it should under the influence of gravitation. There seemed only one solution, and this was to assume that there must be yet another planet, farther out, affecting Uranus by its own gravitational pull. To calculate where such a planet should be and how massive it must be to account for the disturbances suffered by Uranus was an enormously difficult task. However, two mathematicians tackled the problem – Urbain Leverrier in France and John Adams in England. Both were successful and both obtained the same result: in 1846 observations were made of the place in the sky where they had predicted another planet would be seen and the planet Neptune was discovered.

After some years it was realized that this planet, too, showed irregularities in its orbit. But by then Newton's basic theory was too well established to be cast aside, and yet another planet was computed to be the cause. The computations followed the lines of those of Adams and Leverrier and were mostly carried out by the American astronomer Percival Lowell, who searched for the planet from his observatory in Arizona. But Lowell had no success with his observations and, when he died in 1916, the quest for the planet was dropped for a time. In 1930 Clyde Tombaugh found it close to the calculated position.

Pluto's discovery seemed another proof of Newtonian principles. But there was still an additional case of difficulty, and this was the behavior of Mercury. The rather oval orbit of this planet was found to rotate, but at a speed greater than that which Newtonian gravitation would lead one to expect. This fact had been appreciated by Leverrier, and he computed an orbit for a disturbing planet nearer to the Sun than Mercury. He announced his results in September 1859 and showed that the effects on Mercury's orbit could be caused by a planet of Mercury's size orbiting the Sun at half its distance. Soon afterward, an amateur astronomer claimed that he had already observed the new planet. Leverrier, who duplicated these results, claimed that the observations were the proof he needed. He named the planet Vulcan.

Though subsequent claims of successful observations were made by professional astronomers, a careful check in March 1877 and October 1882, when Vulcan should have been observed moving across the face of the Sun, showed nothing. Leverrier's conclusion about Vulcan could no longer be accepted. It remained for Einstein's 1916 theory of relativity to account almost exactly for Mercury's orbit.

21. According to the passage, the validity of Newton's gravitational theory was supported by which of the following?

    I. The discovery of the cause of the irregularity of Uranus' orbit
    II. The final explanation of unexpected characteristics of Mercury's orbit
    III. The presence of Pluto as predicted by mathematical computations

    (A) I only (B) II only (C) I and II only
    (D) I and III only (E) I, II, and III

22. According to the passage, John Adams predicted which of the following?

    (A) The existence of Uranus
    (B) The location of Neptune
    (C) The rotation of Pluto
    (D) The location of Vulcan
    (E) The oval orbit of Mercury

23. The author implies that reported sightings of Vulcan were most likely the result of

    (A) insufficient thoroughness in interpreting evidence
    (B) inadequate instructions provided by Leverrier's theory
    (C) interference from the planet Mercury, which eclipsed Vulcan
    (D) dominance of the field by amateur astronomers
    (E) misinterpretations of data that Newton had collected

24. According to the passage, the implications of Newton's theory of gravitation were adequate to account for which of the following?

    I. Neptune's orbit
    II. Pluto's orbit
    III. Mercury's orbit

    (A) I only (B) II only (C) III only
    (D) I and II only (E) I, II, and III

25. Which of the following titles is most suitable for the passage?

    (A) How Planets Are Sighted
    (B) Why Newton Was Wrong
    (C) Newton's Gravitational Theory and the Discovery of Planets
    (D) Development of Astronomical Theories in the Nineteenth Century
    (E) The Role of Mathematics in Astronomy

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Law and morality are closely related, and interact in a highly complex way; moreover, there always exists the possibility of serious divergence in a given situation between the duty imposed by law and that imposed by morality. There are three main attitudes that may be adopted toward the possibility of such a divergence, and these are encapsulated here.

First, it may be said that law and morality must necessarily coincide either because morality dictates the actual content of human law, as in the case of the Hebrew and Calvinist theocracies, or alternatively because morality is itself merely what the law lays down. The first alternative leads to the proposition that only morality is valid and that nothing that does not conform to it can be regarded as binding. The second alternative has been propounded by many eminent philosophers. Hobbes, for instance, argued that morality itself means nothing more than obeying the law, so that "unjust law" is a contradiction in terms. Hegel's mystical theory of the superiority of the state over the individual proposed that the individual could claim no higher right than to obey the law of the state of which he or she forms an insignificant part.

The second attitude asserts that law and morality are distinct, but that morality provides a touchstone for the validity of merely human law. Conflicts therefore need to be resolved, at least in the last resort, in favor of morality. Broadly speaking this approach to morality has, beginning with the Greeks, been discussed in Western jurisprudence mostly in terms of a so-called law of nature, which is regarded as containing precepts of a higher order, whether divinely ordained or part of the material world. This doctrine has been linked with the idea of the "natural rights" of the individual, which has played so large a role in democratic thought since the American and the French revolutions.

Third, there is the approach that treats the autonomy of law and morality as exclusive, so that neither can resolve questions of validity save in its own sphere. This theory is commonly referred to as "legal positivism." It insists that the validity of a legal rule can depend solely on legal criteria, just as moral validity depends on such criteria as are necessary or appropriate to a system of morality. Those who support this view hold that any conflict between the two spheres cannot impugn the validity of human law or alter the duty of legal obedience, though it may give rise to the moral problem of whether the law ought to be changed. And, in extreme cases, a conflict between legal and moral duty may have to be resolved in accordance with the conscience of individuals and their moral courage to defy a law that they believe to be contrary to what is morally right or just.

26. It can be inferred that throughout the passage the term "human law" refers to law that is

(A) established by legislative agents
(B) conceived in the spirit of compassion
(C) comprehensible to most lay people
(D) meant to govern all behavior
(E) hostile to spiritual concerns

27. It can be inferred from the passage that in a theocracy the law is

(A) the exclusive domain of priests
(B) the ultimate mystery of faith
(C) a matter of relative unimportance
(D) a reflection of ethical precepts
(E) a precondition of advanced spirituality

28. It can be inferred that "the idea of the 'natural rights' of the individual" (lines 35-36) plays a great role in revolutionary democratic thought because that idea

(A) forces one to disregard all traditional restraints
(B) calls for the abolition of religious institutions
(C) attacks the validity of all nonscientific social theories
(D) implies that anarchy is humanity's native condition
(E) justifies the infraction of repressive laws

29. It can be inferred from the passage that a legal positivist would consider which of the following to be logically possible?

I. An act that is morally reprehensible but legally correct
II. An act that is morally correct but legally reprehensible
III. An act that is morally reprehensible and legally reprehensible

(A) II only
(B) III only
(C) I and III only
(D) II and III only
(E) I, II, and III

30. According to the passage, "in extreme cases" (line 51) legal positivism allows for the possibility of

(A) irrational behavior
(B) altruistic sacrifice
(C) inspiring leadership
(D) civil disobedience
(E) religious fanaticism

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6

Select the word or set of words that best completes each of the following sentences.

31. Recent legislative action will ---- the Pine Barrens from random and harmful growth while conversely ensuring that orderly and sound commercial development will not be ----.

(A) protect. .halted (B) remove. .slowed
(C) defend. .continued (D) isolate. .undertaken
(E) preserve. .allowed

32. As a young man, Jean Toomer spoke of his desire to integrate the seemingly ---- elements of his racial heritage so that they might ----, rather than oppose, each other.

(A) archaic. .reprove (B) contrary. .complement
(C) dissimilar. .dispel (D) baffling. .exceed
(E) inverted. .induce

33. Mary Cassatt, an Impressionist painter, was the epitome of the ---- American: a native of Philadelphia who lived most of her life in Paris.

(A) conservative (B) provincial
(C) benevolent (D) prophetic
(E) expatriate

34. Long after science has recognized that appearances are relative, many people nevertheless continue to consider them ----.

(A) illusive (B) uncertain (C) absolute
(D) dependent (E) exceptional

35. Because it has always ---- so many of life's pleasures, this sect finds it difficult to ---- its membership in today's hedonistic atmosphere.

(A) endorsed. .placate
(B) forbidden. .deplete
(C) proscribed. .augment
(D) denounced. .alienate
(E) condoned. .supplement

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. CLOCK : MINUTE :: (A) ruler : centimeter
(B) sundial : shadow (C) arc : ellipse
(D) product : shelf life (E) quart : capacity

37. LAWYER : CLIENT ::
(A) doctor : surgeon
(B) admiral : sailor
(C) judge : defendant
(D) musician : audience
(E) tutor : student

38. CHILL : COLD :: (A) parch : dry
(B) crush : soft (C) freeze : white
(D) graze : opaque (E) scrub : hard

39. RECUPERATE : HEALTH ::
(A) repent : wickedness
(B) respond : medication
(C) regret : opportunity
(D) revive : consciousness
(E) revenge : assault

40. LEGEND : MAP :: (A) volume : manual
(B) career : biography (C) glossary : text
(D) column : article (E) issue : print

41. RIFLE : MUNITIONS ::
(A) band : instruments
(B) cash : merchandise
(C) tool : measurement
(D) spatula : utensils
(E) dumbbell : calisthenics

42. CONFLAGRATION : FIRE ::
(A) precaution : danger (B) jubilee : celebration
(C) challenge : perfection (D) defense : attack
(E) ornamentation : communication

43. HIATUS : ACTIVITY :: (A) link : chain
(B) day : eternity (C) mesh : net
(D) intermission : play (E) revision : edition

44. LIONIZE : CELEBRITY :: (A) tyrannize : ruler
(B) ostracize : outcast (C) demoralize : censor
(D) idolize : worshipper (E) familiarize : guide

45. CHARISMATIC : LOYALTY ::
(A) crass : sophistication
(B) odious : repugnance
(C) famous : superiority
(D) pessimistic : expectation
(E) melodramatic : contradiction

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

Correct Answers for Scholastic Aptitude Test Form Code 4I

VERBAL		MATHEMATICAL	
Section 1	Section 6	Section 3	Section 5
1. B	1. C	1. C	1. B
2. A	2. D	2. E	2. B
3. B	3. A	3. E	3. B
4. E	4. D	4. D	4. D
5. D	5. E	5. B	5. D
6. C	6. E	6. B	6. A
7. B	7. C	7. C	7. D
8. A	8. E	8. D	*8. A
9. B	9. D	9. A	*9. B
10. C	10. B	10. B	*10. C
11. A	11. D	11. D	*11. D
12. C	12. E	12. A	*12. B
13. E	13. B	13. D	*13. A
14. B	14. E	14. B	*14. C
15. D	15. B	15. C	*15. B
16. D	16. A	16. B	*16. D
17. A	17. C	17. A	*17. C
18. E	18. C	18. D	*18. B
19. D	19. E	19. E	*19. C
20. B	20. C	20. B	*20. A
21. D	21. D	21. E	*21. D
22. C	22. B	22. E	*22. C
23. A	23. A	23. A	*23. D
24. A	24. D	24. E	*24. A
25. E	25. C	25. A	*25. D
26. C	26. A		*26. D
27. C	27. D		*27. A
28. E	28. E		28. D
29. B	29. E		29. C
30. B	30. D		30. D
31. E	31. A		31. C
32. D	32. B		32. B
33. C	33. E		33. B
34. D	34. C		34. E
35. A	35. C		35. E
36. A	36. A		
37. A	37. E		
38. D	38. A		
39. E	39. D		
40. B	40. C		
	41. D		
	42. B		
	43. D		
	44. B		
	45. B		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

SAT-Verbal Sections 1 and 6

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 6* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 94. Enter this figure on the worksheet.

SAT-Mathematical Sections 3 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 94. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 6: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 94.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 3: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 94.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 4I

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	600
84	780		39	450	590
83	760		38	450	580
82	750		37	440	570
81	740		36	430	570
80	730		35	430	560
79	730		34	420	550
78	720		33	410	540
77	710		32	410	530
76	700		31	400	520
75	690		30	390	510
74	690		29	390	500
73	680		28	380	500
72	670		27	370	490
71	660		26	370	480
70	660		25	360	470
69	650		24	350	460
68	640		23	350	450
67	630		22	340	440
66	630		21	330	440
65	620		20	330	430
64	610		19	320	420
63	610		18	310	410
62	600		17	310	400
61	590		16	300	390
60	590	800	15	290	380
59	580	780	14	290	380
58	570	770	13	280	370
57	570	760	12	270	360
56	560	750	11	270	350
55	550	740	10	260	340
54	550	730	9	260	340
53	540	720	8	250	330
52	540	710	7	240	320
51	530	700	6	240	310
50	520	690	5	230	300
49	520	680	4	220	300
48	510	670	3	220	290
47	500	660	2	210	280
46	500	650	1	200	270
45	490	640	0	200	270
44	490	640	−1	200	260
43	480	630	−2	200	250
42	470	620	−3	200	250
41	470	610	−4	200	240
			−5	200	230
			−6	200	220
			−7	200	220
			−8	200	210
			−9	200	200
			or below		